



Examining the Crime Concentration: Spatial Patterns of Burglary Crime in Taipei, Taiwan

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Abstract

Research shows that crime is spatially concentrated. However, most research on crime concentration has been conducted in U.S. or European cities. This study assessed whether the "law of crime concentration at place" applies to non-Western settings. This study intends to extend the law of crime concentration's applicability to a different urban and cultural settings. I tested this by focusing on burglary incidents in Taipei City, Taiwan, from 2015 to 2019. I used data from the Taipei City Police Department, focusing exclusively on burglary over 5 years. I employed descriptive statistics, Lorenz curves, and Gini coefficients to quantify and visualize crime concentration at the micro-place level. The units of analysis are clusters of 90 housing units, similar to street segments used in other studies. Consistent with the law of crime concentration at place, burglary in Taipei City is significantly concentrated. Specifically, 51.3% of burglaries occurred in just 5% of micro-place units and 25.9% in only 1.7% of units. Over 80% of spatial units were burglary-free. The generalized Gini coefficient of 0.4967 indicates a noteworthy disparity in the distribution of burglary incidents across different micro-places. I found evidence of crime concentration in a non-Western setting, suggesting that similar patterns of crime distribution may exist across different cultural and geographic contexts. However, socio-economic factors such as residential density and poverty may influence the lower burglary rates in Taiwan than in the U.S. The study site is a densely populated urban setting that may not represent other regions or rural areas in East Asia. Further research is needed to explore the applicability of the law of crime concentration in diverse geographic and socio-economic contexts.

Keywords Crime concentration · Spatial analysis · Burglary · Property crime

Introduction

The "law of crime concentration at place," proposed by Weisburd (2015, p. 138), posits that a disproportionate amount of crime is concentrated in small geographic units. The concentration of crime has been tested across several cities worldwide, including Vancouver, Belgium, Hague, and Nigeria (Lee et al., 2017; also see Andresen et al., 2017a, 2017b; Hardyns et al., 2019; Steenbeek & Weisburd, 2016; Umar et al., 2021; Vandeviver & Steenbeek, 2019). This substantial body of research on geographic concentration of crime suggests that variation in environmental features might potentially play an important role in criminal activities. The findings also imply that these crime hot spots

should be where law enforcement implements interventions, as indicated in numerous empirical studies on hot spot policing (Braga & Bond, 2008; Braga & Weisburd, 2020; Braga et al., 2014; Koper et al., 2021; Sherman & Weisburd, 1995; Weisburd & Telep, 2014a, 2014b). For example, Braga et al. (2014) found that when the police focus on these crime hot spots, they can significantly reduce the crime rate. Although the empirical phenomenon that crime is highly concentrated in places has been central to the study of crime and place, the majority of studies were conducted in Western settings, with limited research exploring whether the "law of crime concentration at place" applies to non-Western contexts, particularly in East Asia. Despite some research efforts in East Asia, studies have yet to test the law of crime concentration at a micro-geographic level due to data limitations (Amemiya & Ohyama, 2019; Liu, 2013; Tsai, 2008). These limitations primarily stem from the fact that available crime data are often aggregated by larger administrative units, such

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as census enumeration districts, districts, or villages, rather than at micro-geographic levels.

In this paper, I examined burglary concentration at the micro-geographic level in Taipei City to fill these gaps in the crime and place literature. By doing so, I aim to test whether the "law of crime concentration at place" holds in an East Asian urban environment, where socio-economic and cultural factors may differ from Western contexts. I used an open-source spatial point data set of burglary crimes (<https://data.taipei/>) and analyzed spatial point patterns over 5 years. Despite a yearly decline in burglary during the study period in the study site, I found that burglary remains concentrated in a small number of micro-place units. The results provide empirical support for crime concentration, indicating that this phenomenon also exists in a non-Western urban setting.

The Law of Crime Concentration at Places

Weisburd (2015, p. 138) proposed a "law of crime concentration at place" in his 2014 Sutherland Address at the American Society of Criminology. He pointed out that "for a defined measure of crime at a specific micro-geographic unit, the concentration of crime will fall within a narrow bandwidth of percentages" (Weisburd, 2015, p. 138). In his research on crime concentration, Weisburd (2015) found that around 1.5% of street segments in eight different U.S. cities accounted for about 25% of the city's crime rate, while about 5% of street segments accounted for half of crime. With an increasing body of research on crime and place, he then argued that this phenomenon could be considered a law of crime concentration and that policing and other crime prevention resources should be targeted in these areas.

Previous research has consistently shown that crime is highly concentrated in specific places (Sherman et al., 1989; Weisburd et al., 2004; Braga et al., 2010, 2011; Gill et al., 2017; Lee et al., 2017; Hardyns et al., 2019; O'Brien, 2019; Walter et al., 2023). For example, Sherman et al. (1989) analyzed approximately 300,000 calls for service data in Minneapolis and found that 50.4% of calls to the police originated from only 3.3% of addresses. This concentration was even more pronounced for specific types of crime. For instance, they found that 2.2% of all places were responsible for 100% of all robbery-related calls to the police (Sherman et al., 1989). Similarly, Weisburd et al. (2004) observed that 5.1% of Seattle's street segments accounted for 50% of the city's crime. Lee and Eck (2014) found that 45.8% of crimes in Cincinnati were concentrated within 5% of the street segments in 2009. Walter et al. (2023) used data from six large U.S. cities from 2008 to 2018 and found that during the study period, 50% of crimes occurred in 3.4% of street segments in San Antonio, 4.7% in Los Angeles, 5.0% in Seattle and New York City, 9.5% in Chicago, and 10.4% in New York City. Furthermore, a systematic review

synthesizing 44 studies showed that crime is concentrated in micro-places regardless of how crime is measured (Lee et al., 2017). These studies provide substantial empirical evidence of crime concentration.

In terms of different crime types, although the degree of concentration varies, crime still tends to be concentrated in a small number of micro-places (Andresen et al., 2017a, 2017b; Braga et al., 2011; Favarin, 2018). For example, Andresen et al. (2017a, 2017b) found that in Vancouver, 50% of assault incidents were concentrated in 1.2% of street segments. For burglary, 50% of incidents were concentrated in approximately 5.1% of street segments, and for robbery, 50% of incidents occurred in 1.1% of street segments. Similarly, Braga et al. (2011) found that nearly 9% of street segments in Boston accounted for over 65% of robbery incidents from 1980 to 2008. When focusing only on commercial robberies in Boston between 1980 and 2008, just 1.3% of street units accounted for nearly half (49.5%) of commercial robberies.

Additionally, over the past 30 years, the key findings of prior research on crime at micro-geographic units revealed that there is crime concentration at places and stability of crime concentration over time regardless of the specific micro-geographic analysis units used in the research (Braga et al., 2010, 2011; O'Brien & Winship, 2017; Sherman et al., 1989; Walter et al., 2023; Weisburd & Amram, 2014; Weisburd et al., 2009). For example, longitudinal research in Seattle (Weisburd et al., 2009) and Tel Aviv (Weisburd & Amram, 2014) has shown that the proportion of street segments producing 50% of the crime remains stable over extended periods. O'Brien and Winship (2017) analyzed over 2 million calls for service and 311 call data in Boston from 2011 to 2013, examining the distribution of crime and disorder. They found a pattern of stability in crime and disorder at the address level during the study years. Additionally, Walter et al. (2023) used data from six large U.S. cities—Chicago, Los Angeles, New York City, Philadelphia, San Antonio, and Seattle—over 11 years and found that crime was significantly concentrated in relatively few street segments, with these street segments displaying a great deal of stability across all cities.

Global Evidence of Crime Concentration

Despite numerous studies reporting significant crime concentration at micro-places in various U.S. cities, similar research did not appear outside the United States until around 2010 (Andresen & Malleson, 2011; Cummings et al., 2022; Favarin, 2018; Steenbeek & Weisburd, 2016; Umar et al., 2021; Vandeviver & Steenbeek, 2019; Weisburd & Amram, 2014). For instance, in Tel Aviv-Jaffa, Israel, approximately 4.5% of street segments accounted for about 50% of crimes in 2010 (Weisburd & Amram, 2014). In

Vancouver, Canada, Andresen and Malleson (2011) found that a small percentage of street segments accounted for a substantial portion of crime. Specifically, their results show that less than 8% of street segments are responsible for 50% of burglaries, and about 1% of street segments are responsible for half of the robberies, and this phenomenon remains stable over time. Favarin (2018) found that the concentration of burglary and robbery was highly evident in Milan, Italy. On average, from 2007 to 2013, 4% of the street segments accounted for 50% of total burglaries, and 1.6% of the street segments accounted for 50% of total robberies. During the study period, robbery consistently exhibited a higher degree of concentration compared to burglary, as confirmed by the Gini coefficients of 0.77 for burglary and 0.87 for robbery (Favarin, 2018). Furthermore, Vandeviver and Steenbeek (2019) focused on burglaries in Antwerp, Belgium and found that a small number of street segments experienced the majority of burglaries, aligning with the law of crime concentration. Their research demonstrated that the percentage of street segments accounting for 50% of burglaries remained relatively stable over time, with about 2% of street segments responsible for half of the burglaries each year from 2005 to 2016.

Even in developing countries like Guyana, robberies were concentrated on 2.8% of Georgetown's streets (Cummings et al., 2022). Cummings et al. (2022) analyzed crime data from 2000 to 2010 in Georgetown, the capital of Guyana, and showed that out of 4,736 street segments, only 131 street segments (2.8%) accounted for all robberies. Furthermore, an even smaller percentage of street segments, 2.0%, were responsible for 50% of all robberies over the 10 years. In Nigeria, similarly, Umar et al. (2021) found that approximately 11.3% of street segments accounted for 50% of crime incidents in 2015.

The Importance of Micro-Place Units in Understanding Crime Concentration

As early as the 1920s, the Chicago School, which focused on neighborhoods, argued that social disorganization explained the uneven spatial distribution of crime and other social and health problems. However, the study of crime and place now emphasizes smaller geographic units, such as street segments. This is because crime concentration is not only existing but also significantly more pronounced in very small specific locations (Weisburd & Telep, 2014a, 2014b). In addition, crime is observed to cluster at the micro-place level and shows variations across different micro-place units within the same neighborhood (Weisburd et al., 2012, 2014). It is implied that some places attract potential offenders more than others, and using higher aggregation levels of data might fail to capture these nuances (Weisburd et al., 2012).

In the study of crime concentration, various types of micro-place units are used, such as addresses, clusters of addresses, block faces, or street segments (Sherman, 1995; Sherman et al., 1989; Taylor, 1997; Weisburd & Green, 1995; Weisburd et al., 2012). According to Eck and Weisburd (1995), a micro-place is a specific location within the larger social environments of communities or neighborhoods. Empirical research proves that smaller units offer critical insights into crime concentration (Lee et al., 2017; Schnell et al., 2017; Steenbeek & Weisburd, 2016). For example, Steenbeek and Weisburd (2016) found that crime was more concentrated in smaller units. Specifically, only 7% of street segments accounted for 50% of crime, compared to 18% for neighborhoods and districts. Similarly, they found that 62% of the variability in crime could be attributed to street segments, while neighborhoods and districts accounted for 6 and 32%, respectively. Schnell et al. (2017) found that 7.2% of street segments accounted for 50% of violent crimes, 14.4–19.7% of neighborhood clusters accounted for 50% of violent crimes, and 18.4–21.4% of community areas accounted for 50% of violent crimes. These empirical findings support the notion that as the size of the place unit decreases, crime concentration increases (Lee et al., 2017).

Furthermore, analyzing crime data at large geographic units such as cities, communities, or census tracts can mask significant geographical variability at lower levels, impacting the validity of findings (Gerell, 2017). Research consistently shows that smaller units of analysis better capture the variance in crime incidents (O'Brien & Winship, 2017; Schnell et al., 2017; Steenbeek & Weisburd, 2016).

The Current Study

The purpose of this study is to extend research on crime concentration by testing the law of crime concentration at micro-places in Taipei City, the capital of Taiwan. Despite a growing body of research on the law of crime concentration in micro-places, very few studies have examined this phenomenon in non-Western urban contexts, specifically at the micro-place units. Additionally, although Amemiya and Oyama (2019) provided an important exploration of crime concentration in Tokyo and Osaka at the district level—with average sizes of approximately 0.195 km² in Tokyo and 0.116 km² in Osaka (Amemiya & Oyama, 2019: p.3)—their analysis used larger spatial units. This limits the applicability of their findings to micro-geographic units, which are crucial in testing the law of crime concentration (Weisburd, 2015; Weisburd et al., 2012, 2014). In contrast, this study aims to address this gap by focusing on micro spatial units, specifically clusters of 90 housing units, to examine burglary concentration. This study is not aiming to directly compare Taipei to other cities; rather, I intend to contribute to comparative criminology by extending the law of crime

concentrations's applicability to a different urban and cultural settings. I used descriptive statistics, Lorenz curves, and Gini coefficients to quantify crime concentration and to increase the understanding of the generalizability of crime concentration in a non-Western urban setting. By examining the spatial patterns of burglary concentration in Taipei City over 5 years, this study contributes to the growing body of literature on crime concentration. It offers valuable insights into focused crime prevention strategies in a densely populated, non-Western urban environment.

Data and Method

Study Setting and Crime Data

This study examined crime trends from 2015 to 2019 in Taipei City, Taiwan. Taipei City is an urban city with a population of approximately 2,600,000 over 105 square miles. It is the 7th most populous urban region in the world. In 2019, Taipei city had an average of 25,000 people per square mile. This dense population is at a similar level as New York City.

Although Taipei City is the economic, political, educational, and cultural center of Taiwan, the population is very homogeneous. In 2019, there were 73,096 foreigners (2.8%) in Taipei city. Most of them are from Vietnam, Indonesia, and the Philippines. Taipei has a relatively low poverty rate (< 1.8%).¹ The number of police officers was 8,054 in 2019. The crime rate of criminal offenses in 2019 was 1,509.26 per 100,000 population,² and the theft rate in 2019 was 212.47 per 100,000 population.³

In Taipei City, larceny offenses are one of the most concerning crimes for the public and a primary focus for crime prevention by law enforcement. In 2019, a total of 5,645 larceny cases were processed, with burglaries accounting for 440 cases (7.8%). Among Taiwan's 21 counties and cities, Taipei's residential burglary rate ranked 9th at 4.4%. Most incidents occurred in commercial districts or areas

with numerous office buildings. The overall clearance rate for larceny offenses was about 88.8% in 2019, with the clearance rate for burglaries specifically being over 90% (Taiwan National Police Agency, Ministry of the Interior, n.d.; Taipei City Police Department, n.d.).

Crime data were publicly available from the Taipei City Police Department (<https://data.taipei/>), focusing exclusively on burglary over a 5-year period from 2015 to 2019. I believe that this 5-year timeframe is sufficient as it aligns with the duration used in previous studies on crime concentration (Andresen & Malleson, 2011; Stanković, 2022). The data, derived from incident reports rather than calls for service, are considered reliable for analysis. Since 2015, the 14 precincts of the Taipei City Police Department have aggregated crime data by date, time, and "address clusters" rather than specific addresses, primarily for confidentiality purposes. Each cluster consists of 30 building numbers (e.g., 1–30, 31–60, 61–90 on a particular street) rather than exact addresses. The original dataset included a total of 2795 burglary incidents from January 2015 to December 2019. After removing missing data, my sample size consisted of 2776 burglary incidents. Throughout this period, the data show a notable decline in residential burglary incidents, dropping from 674 in 2015 to 402 in 2019, culminating in a significant decrease of approximately 40.4 percent over the 5 years.

Units of Analysis

Taipei City comprises 12 districts and 456 villages. The areas of these villages range from 0.2 to 1.4 square miles, with a mean of 0.6 square miles. District areas vary from 5.7 to 62.4 square miles, averaging 22.7 square miles. Given the absence of smaller, suitable units, micro-level analysis using units such as street segments is not feasible. Therefore, to conduct a spatial analysis of crime patterns, I aggregated these address-based burglary data into larger units, which I term "address clusters." This approach was feasible because the original crime data were organized by groups of addresses (e.g., 1–30, 31–60, 61–90 housing units) rather than specific addresses.

In the original dataset, burglary incidents were reported by clusters of 30 building numbers, rather than by exact addresses (e.g., No. 1–30, Ganghua St., Neihu Dist.; No. 121–150, Jilin Rd., Zhongshan Dist.). Given this existing grouping, I further aggregated these clusters by combining every three clusters to form a unit of analysis that mirrors the concept of street segments or addresses commonly used in international studies (Sherman et al., 1989; Weisburd, 2015). In Taipei, households are arranged contiguously, with building numbers typically following a sequential pattern along a street. On one side of the street, buildings are numbered with odd numbers (e.g., 1, 3, 5), while on the opposite side, they are numbered with even numbers (e.g., 2, 4, 6). For

¹ In Taiwan, poverty is defined by the "minimum cost of living," with individuals earning less than 60% of the median disposable income considered to be below the poverty level. Additionally, "low- and middle-income households" are calculated at 1.5 times the minimum living expenses, but cannot exceed the national median.

² Crime rate = (Number of crime incidents / Mid-year population) * 100,000.

³ The theft rate includes general theft (including residential burglary), major theft (cases where the total value of stolen goods exceeds USD \$32,000; theft involving firearms, ammunition, explosives, or important national defense, transportation, and academic facilities or equipment; cases involving foreign diplomats or visiting foreign dignitaries as victims; theft of important instruments or documents affecting national security or social order), and theft of automobiles and motorcycles.

example, the cluster "No. 1–90, Sec. 1, Zhongshe Rd., Shilin Dist." includes buildings on both sides of the street within a continuous section. Thus, each unit of analysis in this study comprises 90 housing units, similar in size and structure to street segments in previous analyses. In simpler terms, each unit of analysis in this study consists of 90 housing units, equivalent to one street segment, covering an average walking distance of 3–5 min. This approach yielded a total of 10,691 units across Taipei City.

Analytic Strategy

I conducted two sets of analyses. Firstly, I presented descriptive statistics of the percentage of spatial units that account for 25 and 50% of crime from the spatial units with the most burglary incidents, as reported in prior crime concentration studies (Braga et al., 2010; Felson et al., 2022; Park & Lum, 2023; Sherman et al., 1989; Weisburd, 2015; Weisburd et al., 2004; Wenger & Lantz, 2022). Although most crime concentration studies reported results using cumulative percentages, there seems to be no accepted standard for selecting specific reference values (different studies select different values, e.g., Braga et al., 2010; Curman et al., 2015; Sherman et al., 1989; Weisburd, 2015). This lack of a unified benchmark complicates comparisons between studies (Hipp & Kim, 2017). Therefore, this study also used the Lorenz curves and Gini coefficient to measure crime concentration (Bernasco & Steenbeek, 2017; Eck et al., 2017; Hipp & Kim, 2017; Steenbeek & Weisburd, 2016).

Using the Lorenz curve and the Gini coefficient provides a visual understanding of crime concentration beyond descriptive statistics alone. The Lorenz curve captures the full extent of how crimes are distributed across different locations. The Gini coefficient quantifies this inequality into a single value, making it easier to compare the degree of concentration across different areas. Additionally, Bernasco and Steenbeek (2017) utilized the Gini coefficient to test crime concentration and suggested that the generalized Gini coefficient might better represent crime concentration in situations with sparse crime data. When the number of crimes approaches and eventually exceeds the number of spatial units, the generalized Lorenz curve and generalized Gini coefficient asymptotically approach the traditional Lorenz curve and Gini coefficient. Thus, regardless the number of crimes in the sample compared to the number of spatial units, the results of the generalized Lorenz and Gini coefficients are comparable across different areas, periods, and crime types (Bernasco & Steenbeek, 2017).

In this study, the Lorenz curve represents the accumulating percentage of crimes on the vertical axis and the accumulating percentage of locations on the horizontal axis, with locations ordered by the number of crimes. Thus, each point on the curve represents data such as "Y percent of crimes

occur in the X percent of locations with the most crimes." Additionally, the information in the Lorenz curve can be quantified using the generalized Gini coefficient of inequality, which is calculated from the Lorenz curve. The formula for the generalized Gini coefficient is as follows (Bernasco & Steenbeek, 2017: p. 459):

$$G' = \max\left(\frac{1}{c}, \frac{1}{n}\right) \left(2 \sum_{i=1}^n (iy_i) - n - 1\right) - \max\left(\frac{n}{c}, 1\right) + 1$$

where, n is the total number of locations (spatial units), c the total number of crime incidents, y_i is the proportion of total crimes that occur at the i -th location, with locations ranked from highest to lowest by crime count, i represents the rank of each location, with 1 being the location with the highest number of crimes (see Appendix A for detailed verification of this formula). This equation uses the generalized Gini coefficient, as proposed by Bernasco and Steenbeek (2017), which adjusts for sparse crime data by employing the line of maximal equality instead of the line of perfect equality.

This formula calculates the generalized Gini coefficient, which takes into account the number of spatial units and crime incidents. Unlike the traditional Gini coefficient, which measures the ratio of the area between the observed Lorenz curve and the line of perfect equality to the area between the line of perfect equality and the line of perfect inequality (Gastwirth, 1972), the generalized Gini coefficient incorporates the line of maximal equality, addressing the potential overestimation of concentration in cases with sparse data (Bernasco & Steenbeek, 2017). The Gini coefficient ranges from 0 (representing perfect equality, where crimes are evenly distributed across all locations) to 1 (representing perfect inequality, where all crimes are concentrated in a single location). In sparse crime data situations, where the number of crimes are less than the number of locations, the generalized Gini coefficient provides a downward adjustment to the traditional Gini coefficient (Bernasco & Steenbeek, 2017). The approach of using the Lorenz curve and Gini coefficient analyzes the trends in crime concentration across different locations and observes how crime concentration changes over time (Johnson, 2010).

Results

Table 1 illustrates the concentration of burglary incidents within Taipei City's street segments, following the approaches taken by Weisburd (2015) (also see, Andresen & Malleson, 2011; Steenbeek & Weisburd, 2016). This method focuses on identifying the percentage of spatial units that account for substantial proportions of crime, a key aspect in the study of crime distribution and concentration.

Table 1 Distribution of Burglary proportions by street segment in Taipei from 2015 to 2019

Burglary incident	Street segment count	Burglary cumulative percent	Segment cumulative percent
30	1	1.1	0.01
9	1	1.4	0.02
8	4	2.6	0.1
7	6	4.1	0.1
6	7	5.6	0.2
5	18	8.8	0.4
4	44	15.2	0.8
3	99	25.9	1.7
2	353	51.3	5.0
1	1352	100	17.6
0	8806		100

Notes. 1. Cumulative Calculation of Street Segments and Segments with No Crime Incidents: The cumulative calculation of "Street Segments" includes only those segments with recorded burglary incidents. The segments with no burglary incidents are presented separately in the final row, ensuring logical consistency and distinguishing between segments with and without crime. In this table, the 8,806 segments with zero incidents are added in the last row to bring the total to 10,691 segments. 2. Method of Calculating Cumulative Percentages: (a) Burglary Cumulative Percent: This column represents the cumulative percentage of burglary incidents, calculated progressively by row. Each value includes the cumulative total from previous rows, building incrementally until it reaches 100%. (b) Segment Cumulative Percent: This column reflects the cumulative percentage of street segments based on the total number of segments (10,691). To maintain consistency, the calculation follows a progressive cumulative approach, and the count of segments with no burglary incidents is displayed separately in the last row, resulting in a cumulative total of 100%.

The findings revealed a significant clustering of crime within a few street segments: 51.3% of all burglaries occurred in just 5.0% of the segments, and a notable 25.9% of burglaries were concentrated in only 1.7% of segments. This pattern of distribution adds empirical evidence to the law of crime concentration at micro-places. Additionally, the fact that over 80% of street segments experienced no burglaries at all highlights the uneven spatial distribution of burglary crime across the city.

Similarly, when examining each year individually, the result still shows the burglary concentration at micro-places. As shown in Table 2, burglary incidents were significantly clustered within a small proportion of street segments each year. For instance, in 2015, 28.8% of burglaries occurred in just 0.3% of the segments, while over 95% of street segments recorded no incidents. In 2016, 28.5% of burglaries concentrated within around 0.8% of segments, again with a substantial portion of segments remaining unaffected by burglary. In 2017, the data reveals that 23.2% of burglaries were clustered in approximately 0.5% of the street segments,

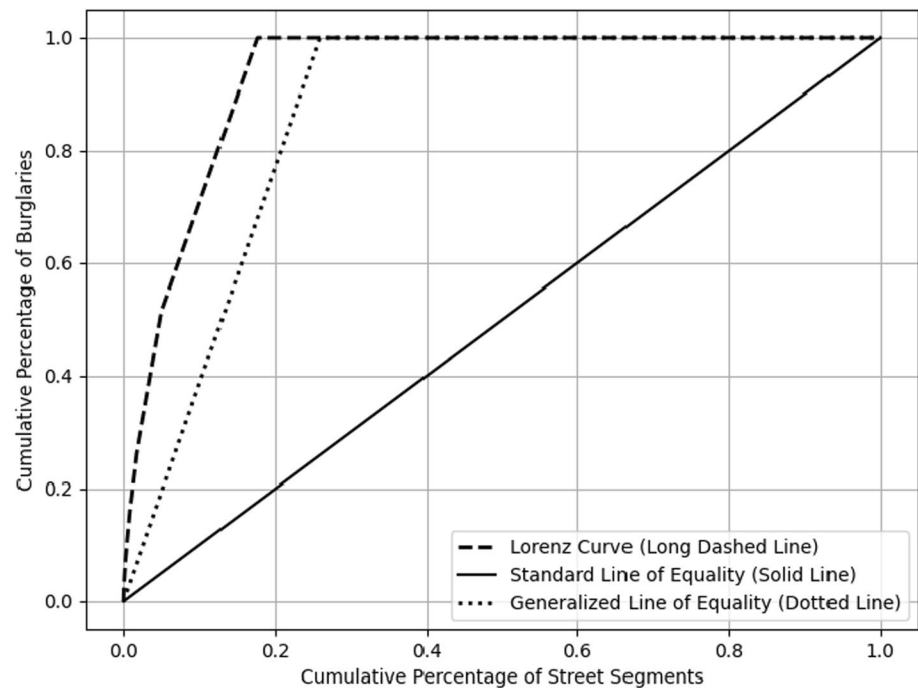
Table 2 Year-by-year distribution of burglary proportions by street segment in Taipei (2015–2019)

Year	Burglary incident	Street segment count	Burglary cumulative percent	Segment cumulative percent
2015	14	1	2.1	0.01
	7	1	3.1	0.02
	5	2	4.6	0.04
	4	4	6.8	0.08
	3	10	11.3	0.2
	2	59	28.8	0.3
	1	479	100	4.8
	0	10,135		100
2016	8	1	1.2	0.01
	5	2	2.7	0.03
	4	2	3.9	0.05
	3	10	8.5	0.15
	2	65	28.5	0.75
	1	465	100	5.05
	0	10,146		100
2017	5	1	0.9	0.01
	4	3	3.1	0.04
	3	11	9.1	0.14
	2	39	23.2	0.54
	1	424	100	4.54
	0	10,213		100
2018	3	6	3.6	0.1
	2	39	19.3	0.5
	1	401	100	4.3
	0	10,245		100
2019	3	5	3.7	0.1
	2	29	18.1	0.4
	1	329	100	3.5
	0	10,328		100

maintaining the trend of high concentration in limited areas while 95% of segments reported no incidents. This trend continued in 2018, where 19.3% of burglaries occurred in just 0.5% of segments, showing a slight decline in concentration yet consistently emphasizing the localization of crime. By 2019, this concentration persists, with 18.1% of burglaries recorded in only 0.4% of segments, while most segments remained free from incidents. This yearly breakdown underscores the persistence of burglary concentration at micro-places, confirming the applicability of the law of crime concentration within Taipei City even when analyzed on an annual basis.

Recognizing the significant clustering of crimes at micro-places, the next step is to examine this pattern through a visualization. The Lorenz curve (Lorenz, 1905) serves as an ideal tool for this purpose, showing how crime is concentrated within limited "bandwidths" or narrow ranges of

Fig. 1 Lorenz curves for Burglary Concentration, Taipei City, 2015–2019



street segments that account for the majority of incidents (Weisburd, 2015).

Figure 1 presents a detailed examination of crime concentration using a Lorenz curve. This curve plots the cumulative percentage of burglary incidents against the cumulative percentage of street segments. Unlike analyses that might require arbitrary cutoff percentages for reporting, the Lorenz curve offers a continuous assessment of crime concentration across varying levels of spatial aggregation. This methodological choice enhances this understanding by allowing interpretations of crime distribution without predetermined thresholds.

Figure 1 reveals a pronounced concentration of burglary incidents within a limited number of street segments. This distributional skewness is quantitatively represented by a Gini coefficient of approximately 0.87, indicating substantial inequality in the distribution of burglary incidents across spatial units. Such a high Gini coefficient suggests that a small proportion of areas bear a disproportionately large share of burglary incidents.

However, when accounting for the sparse nature of the data (my sample size is 2776 burglary incidents during the study period), the generalized Gini coefficient for burglary is 0.4967. This adjustment provides a more accurate representation of crime concentration under conditions with more spatial units than crimes (Bernasco & Steenbeek, 2017). Literature suggests that a generalized Gini coefficient of around 0.5 indicates moderate inequality in the distribution of crime incidents among spatial units (Bernasco & Steenbeek, 2017; Eck et al., 2017; also see Lee, 2017; Mohler et al., 2019). For

instance, Bernasco and Steenbeek (2017) showed that for certain types of crimes, such as assaults, the concentration appeared more moderate (around 0.5 of the generalized Gini coefficient) when the number of spatial units significantly exceeded the number of crime incidents. Thus, the generalized Gini coefficient of 0.4967 in this study underscores a notable disparity in the distribution of burglary incidents, suggesting that specific areas still experience a higher burden of crime despite the overall sparsity.

In addition to the cumulative analysis presented in Figs. 1, 2, 3, 4, 5, and 6 show annual breakdowns of Lorenz curves for burglary incidents across street segments from 2015 to 2019. For 2015, as illustrated in Fig. 2, the Gini coefficient was approximately 0.96, with the generalized Gini coefficient approximately 0.31, indicating a significant concentration of crime within a small number of street segments. This trend of high concentration continues across subsequent years, with minor variations. In 2016 (Fig. 3), the Gini coefficient increased slightly to 0.96, and the generalized Gini coefficient was approximately 0.28, reflecting a similarly uneven distribution.

In 2017, shown in Fig. 4, the Gini coefficient reached approximately 0.96, with the generalized Gini coefficient of 0.24, indicating heightened concentration yet reflecting the sparse crime incidents relative to spatial units. For 2018, displayed in Fig. 5, the Gini coefficient remained high at 0.96, with an adjusted Gini coefficient of 0.19. Finally, in 2019 (Fig. 6), the Gini coefficient reached 0.97, while the generalized Gini coefficient dropped to 0.18. The

Fig. 2 Lorenz curves for Burglary concentration, Taipei City in 2015

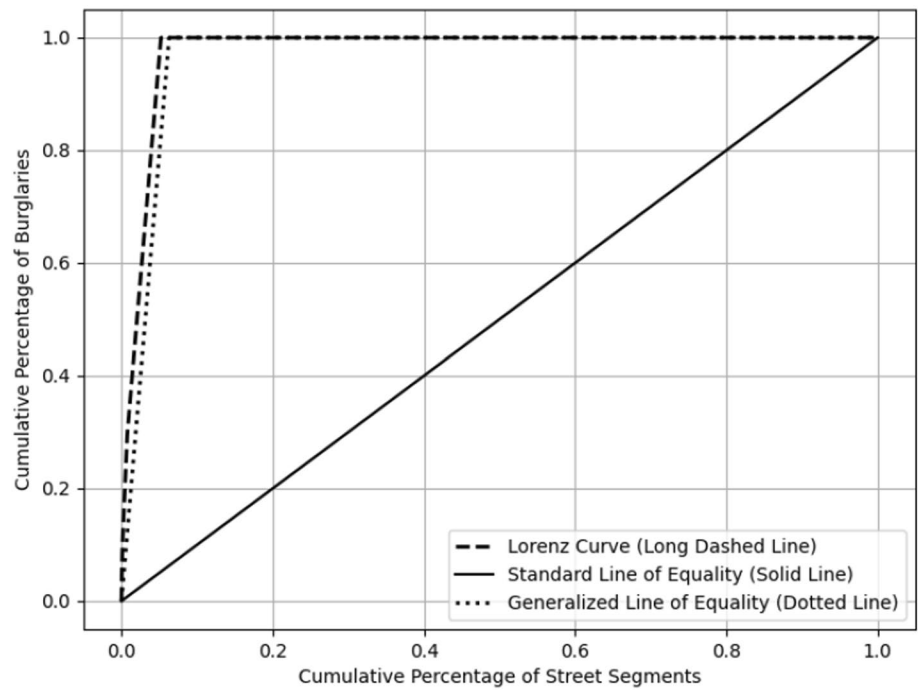
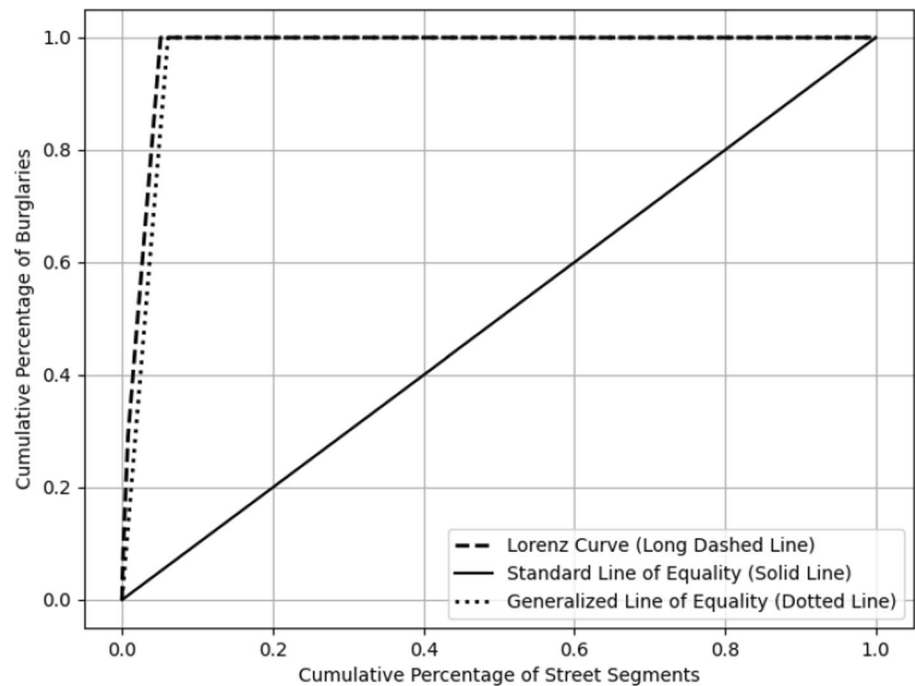


Fig. 3 Lorenz curves for Burglary concentration, Taipei City in 2016



disparity between the Gini coefficient and the generalized Gini coefficient, specifically in 2018 and 2019, may be attributed to a combination of factors related to data sparsity and spatial distribution. As the number of burglary

incidents decreases, the data becomes increasingly sparse. This sparsity may impact the generalized Gini coefficient.

Fig. 4 Lorenz curves for Burglary concentration, Taipei City in 2017

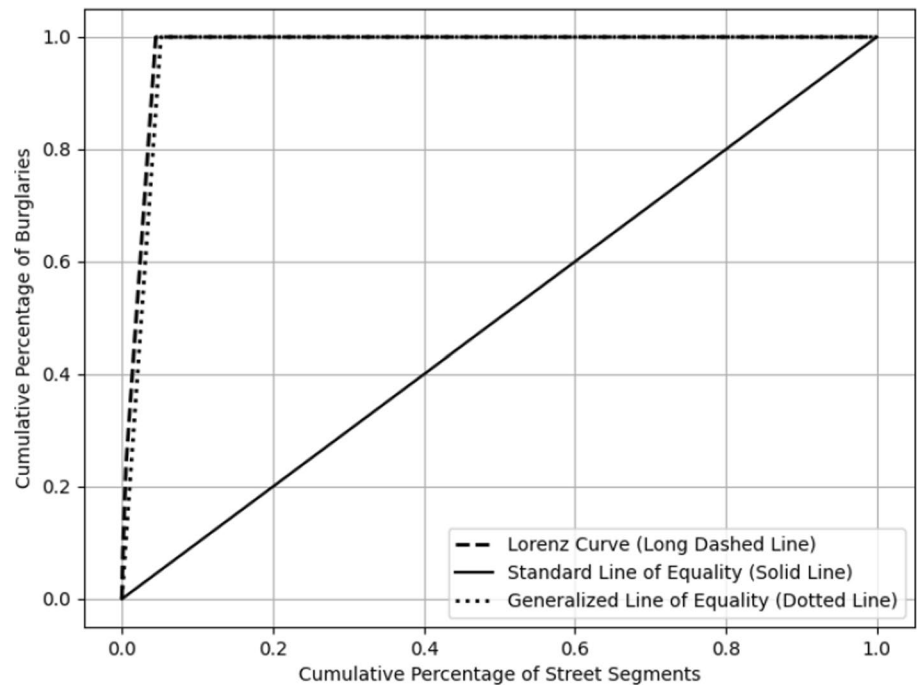
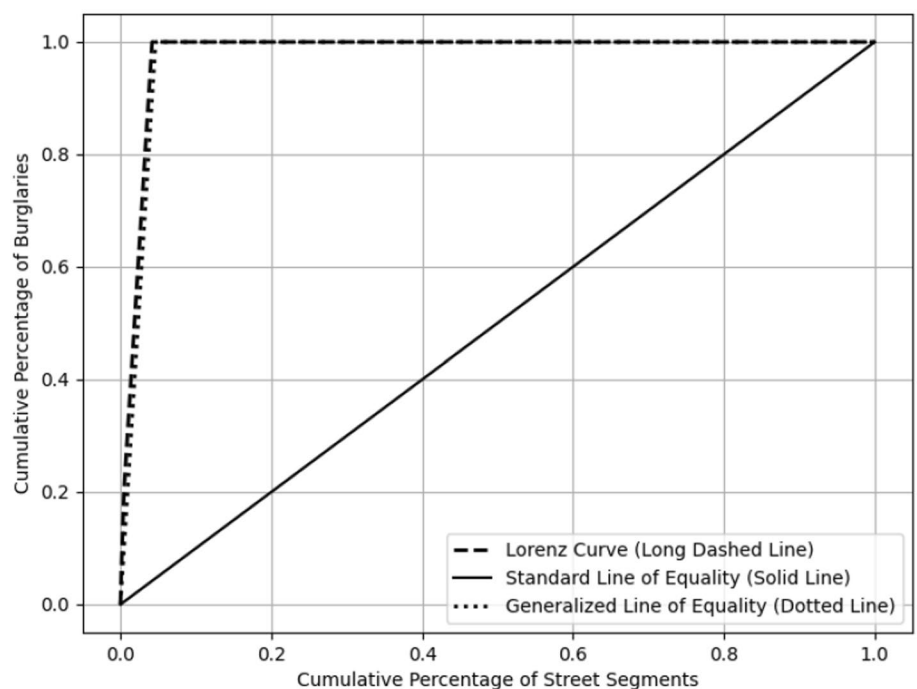


Fig. 5 Lorenz curves for Burglary concentration, Taipei City in 2018

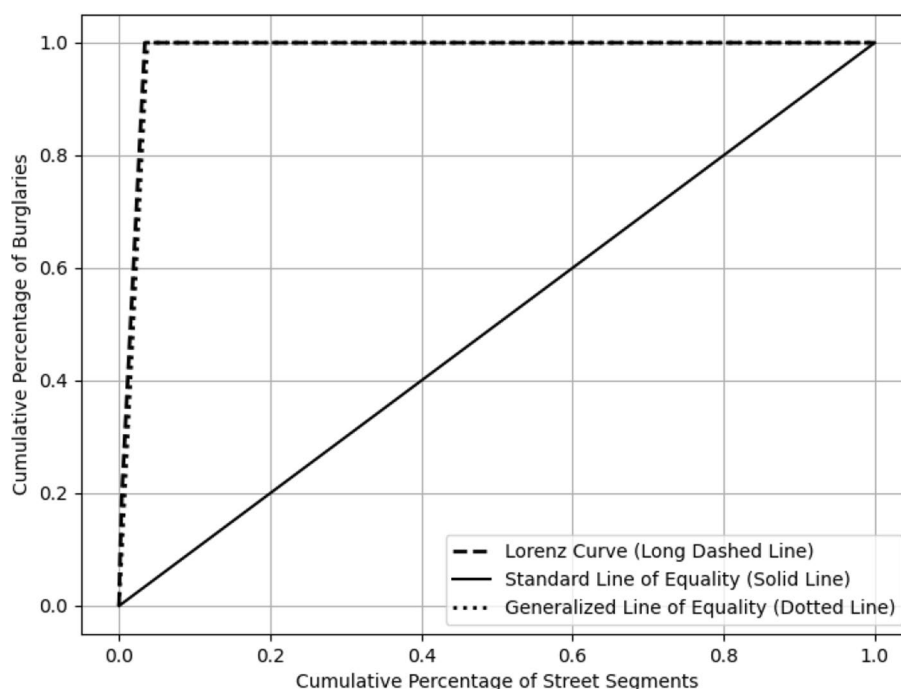


Discussion

In this study, I employed a comprehensive analytical approach, combining descriptive statistics, Lorenz curves, and Gini coefficients to assess the concentration patterns of burglary across address clusters in Taipei City, Taiwan. The findings of this study provide evidence that the "law of

crime concentration at place" is applicable to a non-Western urban context, specifically in Taipei City, Taiwan. This supports the notion that crime is not randomly distributed but instead is significantly concentrated at specific micro-places, aligning with the substantial and growing body of evidence of the patterns observed in U.S. and European cities (Andresen et al., 2017a, 2017b; Favarin, 2018; Gill et al., 2017; Lee et al., 2017; Levin et al., 2017; Sherman

Fig. 6 Lorenz curves for Burglary concentration, Taipei City in 2019



et al., 1989; Weisburd, 2015; Weisburd et al., 2012, 2014), this study is among the few attempts to examine the spatial patterns of crime in a non-Western urban setting, particularly in Taipei, Taiwan (Amemiya & Ohyama, 2019; also see Lee et al., 2017). Additionally, while Amemiya and Ohyama (2019) provided valuable insights into crime concentration in large Japanese cities (Tokyo and Osaka), their study focused on larger spatial units, such as census enumeration districts, limiting the ability to observe crime concentration at the micro-geographic level. Research in the law of crime concentration increasingly emphasizes smaller geographic units, such as street segments, because crime concentration is significantly more pronounced in very specific micro-places (Schnell et al., 2017; Steenbeek & Weisburd, 2016; Weisburd, 2015). Studies have demonstrated that crime clusters vary even within the same neighborhood, indicating that using larger spatial units may obscure the precise patterns of concentration (Weisburd et al., 2012; Steenbeek & Weisburd, 2016; Schnell et al., 2017). By focusing on micro-place units, such as clusters of housing addresses in Taipei, this study better captures the nuanced distribution of crime compared to studies using larger aggregation levels like census districts. This unit of analysis difference provides an advantage in testing the law of crime concentration because smaller units reveal more detailed patterns of where crime occurs (Lee et al., 2017; Steenbeek & Weisburd, 2016). Thus, this study extends the existing body of research on crime concentration by demonstrating that the fundamental

principles underlying the law of crime concentration at micro-places are not confined to Western contexts.

In Taipei City, a small number of micro-places were found to account for a disproportionate share of burglary incidents, with 51.3% of burglaries occurring in just 5% of the analyzed units and 25.9% in only 1.7% of the units during the 5-year study period. This high concentration of crime in specific places is consistent with findings from other cities worldwide (Andresen et al., 2017a, 2017b; Favarin, 2018; Levin et al., 2017; also see Weisburd et al., 2004; Weisburd, 2015), which might imply that variations in environmental and situational factors play a crucial role in determining where crimes are most likely to occur. Unlike Amemiya and Ohyama (2019), who examined crime concentration using larger spatial units, such as census enumeration districts in Japanese cities, this focus on micro-place units offers a more detailed understanding of crime distribution in a non-Western context. This pronounced concentration of burglary incidents in Taipei City underscores the importance of micro-geographic analysis in urban crime studies and has important implications. It highlights the potential for targeted policing and resource allocation to reduce crime effectively in high-concentration areas (Braga et al., 2011).

The generalized Gini coefficient, adapted from the traditional Gini coefficient, is often employed in crime concentration studies to account for the sparse nature of crime data, where many spatial units report zero or minimal incidents (Bernasco & Steenbeek, 2017). In situations with low crime volumes, however, the coefficient can present interpretive challenges (Curiel, 2019; Mohler et al., 2019). Specifically,

as the total number of crime incidents decreases, the generalized Gini coefficient often trends lower, suggesting a shift towards random distribution rather than high concentration (Curiel, 2019; Mohler et al., 2019). This behavior may unintentionally obscure actual concentration patterns since areas with few crimes might still experience clustering due to specific environmental or situational factors.

In this study, for example, if I examine the annual breakdowns, in 2015, with a burglary count of 674, the Gini coefficient was approximately 0.96, reflecting a high concentration of crime incidents within a limited number of street segments. The generalized Gini coefficient for the same year was adjusted down to 0.31, which captures some degree of inequality but suggests a less pronounced concentration. However, by 2019, with only 402 burglaries, the Gini coefficient further increased to 0.97, indicating extreme concentration, while the generalized Gini coefficient decreased to 0.18. This pattern illustrates how, in contexts of sparse crime data, the generalized Gini coefficient can mask significant clustering by interpreting low counts as indicative of random distribution. Thus, while the generalized Gini coefficient provides an important adjustment for data sparsity, its tendency to interpret low crime volumes as random may reduce its sensitivity to true clustering, making it necessary to interpret results carefully in low crime settings.

Furthermore, recent studies analyzing crime concentration using micro-place units report a generalized Gini coefficient of around 0.6 to 0.8 (Eck et al., 2017; Mohler et al., 2019; O'Brien, 2019). In this study, the generalized Gini coefficient for burglary concentration in Taipei City was 0.4967. The similarity between the Gini coefficients in this research and previous studies, although at a slightly lesser degree, supports the conclusion that crime concentration is evident in Taipei City, just as it is in other cities studied, at least at the micro-place level. This difference might be attributed to unique aspects of Taipei's urban landscape, such as higher population density, distinct residential patterns, and different policing strategies. In Taipei, burglary hotspots are commonly found in areas with mixed land use, where the combination of residential and commercial properties creates an environment with higher foot traffic and more complex movement patterns. These conditions might provide more opportunities for offenders to blend in and commit burglaries without raising suspicion (Lin, 2019; Yue et al., 2018). For instance, mixed-use areas often have shops on the ground floors and residential units above, leading to a lack of community cohesion that can make these areas more vulnerable to crime (Lee, 2005). Additionally, limited natural surveillance during certain hours, particularly at night when businesses are closed, may leave residential units relatively isolated, further contributing to higher burglary risks in these locations (Lee, 2005). In contrast, purely residential areas with lower levels of commercial activity

tend to benefit from natural surveillance. Residents in these areas are often more vigilant about unfamiliar individuals entering their neighborhoods, increasing the likelihood of detecting potential offenders (Lee, 2010). A study on burglary patterns in suburban Taipei revealed that residents in less commercialized neighborhoods exhibited heightened vigilance (Wang, 2005).

Additionally, Taipei's street layout, particularly the width of roads and the connectivity between streets, plays a role in influencing burglary patterns. Areas with high street connectivity and roads wider than 10 m are associated with a lower probability of residential burglary, as these layouts tend to enhance visibility and natural surveillance (Huang, 2002; Wang, 2005). However, Jou (1997) found in her spatial analysis of juvenile crime in Taipei that high-traffic roads, despite their connectivity, were more prone to juvenile residential burglary incidents. The variation in the Gini coefficient between Taipei and other cities suggests that while the general principle of crime concentration is broadly applicable, local nuances must be considered when developing and implementing crime prevention strategies.

The results can also be explained by environmental criminology theories suggesting that "crime occurs in particular places and spaces" (Brantingham & Brantingham, 1981, 1995). The presence of crime hotspots in Taipei City can be attributed to the spatial distribution of facilities and activities that attract offenders. For instance, areas with high commercial activities may provide more opportunities for burglaries due to the increasing mobility and transportation of goods—suitable targets. Potential offenders might observe the routines of these targets, including the break time of security guards and when residents leave for work or return home. This concentration of activity provides ample opportunities for offenders to identify vulnerable targets.

While burglary rates are relatively low in Taiwan compared to the U.S. (11.9 cases per 100,000 people vs. 340.5 cases per 100,000 people, respectively), this study's value lies in extending the crime concentration framework to a non-Western setting. My intention is not to conduct comparative criminology *per se* but rather to offer U.S. burglary rates as a familiar reference point, helping to contextualize the situation in Taipei. One possible factor for the low burglary rate in Taipei is the robust security infrastructure (Yeh, 2014). In Taipei City, advanced home security systems such as CCTV, access control systems, and 24-h connections with police stations are prevalent. Additionally, the security industry offers sophisticated alarm systems and monitoring services, which act as a deterrent to potential offenders. Another factor may be Taiwan's high clearance rate for burglary crimes (95.07% in Taiwan in 2019 according to the Ministry of Interior, compared to 14.1% in the U.S. in 2019 according to the UCR report), which likely serves as a significant deterrent as well.

The lower burglary rate in Taipei may also be linked to lower levels of social disorganization compared to Western contexts (Wang, 2015; Yang & Hoffmann, 1998). As outlined in social disorganization theory (Shaw & McKay, 1969; Kubrin & Weitzer, 2003), factors such as high residential mobility, ethnic heterogeneity, and weak social ties contribute to higher crime rates in communities. In Taiwan, community ties and social cohesion, often shaped by family and neighborhood relationships, are stronger due to cultural and social norms (Lin et al., 2010). These ties foster higher levels of informal social control, particularly in densely populated areas like Taipei (Shen, 1987; Tsai, 1987). Unlike in Western cities, where suburban sprawl often increases opportunities for burglary (Brantingham & Brantingham, 1984; Hipp & Kim, 2017), Taipei's high residential density and close-knit social networks provide additional layers of informal surveillance, reducing crime opportunities (Lin, 2019). The overlapping relationships between kinship, neighbors, and classmates also play a critical role in reinforcing community ties and further deterring crime (Fei et al., 1992; Lin et al., 2010).

Additionally, considering that residential burglaries are more likely to occur in low-density areas because they are easy to enter and there are usually multiple escape routes (Nee & Taylor, 2000), the high residential and population densities in Taipei City make it challenging for potential offenders to find multiple escape routes, further reducing burglary opportunities (Lin, 2019). Specifically, the proximity of residences and the potential for increased surveillance by neighbors can deter burglars, leading to a relatively higher concentration of burglaries compared to previous studies conducted in Europe and the United States.

The findings align with the broader literature on crime concentration and provide valuable insights into the specific context of Taipei City. The significant concentration of burglary incidents in a small number of street segments suggests that targeted policing and community interventions in these hotspots could potentially be effective. These study's results also contribute to the ongoing discourse on the applicability of the crime concentration phenomenon across different cultural and micro-geographic settings. Future research should continue to explore the dynamics of crime concentration over time and across different types of crime to understand further the underlying factors contributing to this phenomenon in Asia.

Another avenue for future research is to explore the role of the integration of advanced spatial analysis techniques and geographic information systems (GIS) to enhance the understanding of crime concentration. Techniques such as spatial autocorrelation, hot spot analysis, and predictive modeling can provide more nuanced insights into the spatial dynamics of crime (Chainey & Ratcliffe, 2013). These

tools can help identify emerging crime hotspots, predict future trends, and inform proactive policing strategies. Furthermore, using big data and machine learning algorithms can facilitate the analysis of large datasets, allowing for more comprehensive and accurate assessments of crime concentration patterns (Mohler et al., 2015).

Implications and Limitations

This study's findings on the concentration of burglaries in specific micro-places in Taipei City support the potential effectiveness of hot spot policing in a non-Western urban environment. Research demonstrates the effectiveness of concentrating police efforts on high-crime places, or "hot spots," to reduce overall crime (Braga et al., 2012, 2014, 2019; Weisburd et al., 2017). Applying these approaches to Taipei could provide similarly efficient burglary reduction benefits but may require adaptations to suit the city's unique urban characteristics.

In Taipei, implementing hot spot policing could involve allocating more resources to identified high-crime locations. Enhanced surveillance through increased patrols, targeted interventions, and the strategic deployment of CCTV cameras could help reduce crime at these hot spots (Weisburd et al., 2003). However, given Taipei's dense, mixed-use environment where residential, commercial, and public spaces often overlap, targeted interventions could also include CCTV deployment and patrol adjustments that account for high foot traffic areas during peak times (Lin, 2019; Yue et al., 2018). Moreover, focusing on specific micro-places, such as high-density residential-commercial zones, could allow law enforcement to address crime more precisely within areas where the concentration of activity and people create unique vulnerabilities (Yue et al., 2018). This adaptation aligns with the Western evidence supporting place-oriented interventions while considering Taipei's specific urban characteristics.

Research shows that hot spot policing has also been associated with a "diffusion of benefits," where crime control effects extend to surrounding areas rather than causing displacement of criminal activity (Braga et al., 2019; Clarke & Weisburd, 1994). Applying hot spot policing in Taipei could yield similar effects, potentially leading to broader community safety improvements beyond targeted high-crime areas. This "spillover" effect could be particularly beneficial in Taipei's mixed-use urban environment, where commercial, residential, and public spaces are closely interconnected.

Additionally, studies show that focused policing strategies can reduce crime without simply shifting it to nearby areas (Braga & Weisburd, 2010; Sherman & Weisburd, 1995). According to Koper (1995), police stops of

11–15 min at high-crime locations are especially effective, producing a lasting deterrence effect even after officers have left. This approach, which has been found more impactful than brief drive-throughs or shorter stops, suggests that training officers in these targeted methods could enhance their ability to allocate resources efficiently and strategically time their presence. For Taipei, where burglary is concentrated in specific micro-places, implementing such patrol strategies could allow law enforcement to focus on these high-risk areas, maximizing deterrence and potentially reducing burglary rates by concentrating efforts where they are needed most (Koper, 1995; Sherman & Weisburd, 1995).

Community-based interventions could further support hot spots policing. By fostering a sense of collective efficacy (such as through neighborhood watch programs) and encouraging community members to take an active role in safeguarding their neighborhoods, even at a micro-place level, the overall crime rates can be reduced (Lanfear, 2022; Sampson et al., 1997; Weisburd et al., 2020). Engaging local residents in crime prevention efforts, such as community patrols, local safety meetings, neighborhood watch programs, and collaborative efforts with local law enforcement, can enhance surveillance, increase informal social control, and strengthen social cohesion, making it more difficult for crimes to go unnoticed and further help reduce opportunities for crime (Gearhart, 2022). This approach could be particularly effective in Taipei, where strong social ties are culturally valued and further facilitate informal monitoring.

Urban planners and policymakers can focus on designing safer urban environments. Implementing principles of Crime Prevention Through Environmental Design (CPTED) can mitigate the risk factors associated with high-crime micro-places (Cozens & Tarca, 2016; Shariati, 2022). Measures such as improved lighting, enhanced visibility, surveillance and physical security, and controlled access to buildings can deter potential offenders and make it more difficult for crimes to occur (Armitage, 2006, 2018; Reynald, 2015; Welsh & Farrington, 2009). For example, ensuring that streets and alleys are well-lit at night and removing obstructions that block the view of public spaces can reduce the opportunities for potential offenders. Additionally, this study's findings have implications for law enforcement training. Understanding the principles of crime concentration and the environmental factors might help increase the effectiveness of proactive policing and crime prevention (Koper, 1995). Law enforcement officers with this knowledge might be more likely to adopt tailored strategies to the specific conditions of their patrol areas.

Furthermore, the findings of this study highlight the need for context-specific crime prevention strategies, particularly in Taipei's densely populated environment. Due to the high volume of people, traditional police patrols may face

limitations. Instead, innovative use of technology, such as CCTV and smart entry recognition systems, can effectively supplement traditional surveillance, enhancing monitoring in ways tailored to Taipei's urban landscape.

Though this study provided insights into place-based criminology, this study has some limitations. These limitations highlight the need for caution in generalizing the findings and suggest directions for future research. The study focuses exclusively on Taipei City, a densely populated urban area. Unlike Amemiya and Ohyama (2019), one of the few studies discussing crime concentration in an East Asian setting, which compared crime concentration across two Japanese cities (Tokyo and Osaka), this study is restricted to one city due to data limitations. The findings may differ from other regions of Taiwan or East Asia, mainly rural or less densely populated areas. The unique urban characteristics of Taipei City, such as its high residential density and advanced security infrastructure, may not be the case in other contexts, limiting the generalizability of the results. Future research could expand this analysis to other cities in Taiwan or East Asia to provide a more comparative perspective.

Unlike typical street segment studies in the U.S., which use precise segments between intersections, I aggregated data based on address clusters comprising 90 housing units. This difference in unit size may influence how burglary patterns are interpreted. Although the consistent sequential numbering of housing units within each cluster allows the unit of analysis to approximate a cohesive spatial area, the size of each cluster may vary relative to traditional street segments. In some areas, a cluster may encompass the equivalent of approximately 1.5 street segments, while in others, it may cover less than one segment. Consequently, some variation in burglary rates within clusters may not be fully captured. While this clustering approach aligns with the data structure, it may introduce a minor averaging effect, which should be considered when interpreting the spatial patterns observed in the results.

Another main limitation of this study, compared to other research on crime concentration in East Asian settings such as Amemiya and Ohyama (2019), is the types of crimes analyzed. Amemiya and Ohyama (2019) examined multiple types of property crimes and found that crime concentration varied by crime type, with commercial burglary being the most concentrated and theft from vehicles more dispersed. This study, however, focuses exclusively on burglary incidents due to data limitations. The findings of this study might differ from those observed for other types of property crimes, and future research could explore whether similar patterns of crime concentration are evident in other crime categories.

While the study provides a descriptive analysis of crime concentration, it lacks data on the environmental features, routine activities, and socio-economic factors at the

identified hotspots. Understanding the factors contributing to crime concentration in these micro-places requires further research. Future studies should focus more on environmental factors and consider incorporating socio-economic variables (such as residential density and poverty rates) and exploring their interaction with environmental and situational factors to provide a more comprehensive understanding of the underlying causes. The study covers 5 years from 2015 to 2019, which may not capture longer-term trends and fluctuations in crime patterns. Future research should consider extended observation periods to assess the stability and dynamics of crime concentration over time.

Conclusion

This study provides support for the law of crime concentration at places in a non-Western urban context using micro-place units. The analysis reveals a high level of burglary concentration in a small number of micro-geographic places in Taipei City, Taiwan. This finding is consistent with previous longitudinal research conducted in the United States, extending the evidence for the "law of crime concentration at place" to a non-Western context. Despite the unique characteristics of this Taipei urban environment, this study demonstrates that the principles of crime concentration are likely applicable across different cultural and environmental contexts. Overall, this study contributes to the growing literature on crime concentration, highlighting the importance of targeted interventions at high-crime places such as hotspot policing or neighborhood watch programs.

Appendix A: Verification of the Generalized Gini Coefficient Formula

The generalized Gini coefficient is an extension of the traditional Gini index, designed to measure crime concentration while addressing potential biases in cases where the number of spatial units exceeds the number of crime incidents. It is defined as follows (Bernasco & Steenbeek, 2017, Eq. 5, p. 459):

$$G' = \max\left(\frac{1}{c}, \frac{1}{n}\right) \left(2 \sum_{i=1}^n (iy_i) - n - 1 \right) - \max\left(\frac{n}{c}, 1\right) + 1$$

where n is the total number of spatial units, c is the total number of crime incidents, y_i represents the proportion of total crimes occurring at the i -th ranked location, i means the position of each place in the ranking, with $i=1$ being the location with the highest number of crimes.

In the manuscript, I used the following formula:

$$G' = \max\left(\frac{1}{c}, \frac{1}{n}\right) \left(2 \sum_{i=1}^n (iy_i) - n - 1 \right) - \max\left(\frac{n}{c}, 1\right) + 1$$

While these formulas appear identical, I proceed with an algebraic verification to ensure absolute consistency.

To simplify the expression and demonstrate its consistency, I use the following identity:

$$\max\left(\frac{1}{c}, \frac{1}{n}\right) = \frac{\max(n/c, 1)}{n}$$

(a) When $n/c \geq 1$, $\max(n/c, 1) = n/c$, then

$$\frac{\max(n/c, 1)}{n} = \frac{n/c}{n} = \frac{1}{c}$$

Since $n/c \geq 1$, which implies that $1/c \leq 1/n$, it follows that $\max(1/c, 1/n) = 1/c$

(b) When $n/c < 1$, $\max(n/c, 1) = 1$, then

$$\frac{\max(n/c, 1)}{n} = \frac{1}{n}$$

Since $n/c < 1$, which implies that $1/c > 1/n$, it follows that $\max(1/c, 1/n) = 1/n$

Therefore, the identity holds in all cases.

Using the above identity in the original formula, I obtain the generalized Gini coefficient as:

$$G' = \frac{\max(n/c, 1)}{n} \left(2 \sum_{i=1}^n (iy_i) - n - 1 \right) - \max(n/c, 1) + 1$$

Knowing that the traditional Gini index is defined as:

$$G = \frac{1}{n} \left(2 \sum_{i=1}^n (iy_i) - n - 1 \right)$$

The equation simplifies to:

$$G' = \max(n/c, 1)G + 1 - \max(n/c, 1)$$

This is precisely the expression derived by Bernasco and Steenbeek (2017).

Declaration

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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